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Independent and joint associations of physical activity and sleep with ADHD in a population-based sample of children: A cross-sectional study

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A B S T R A C T

Background: This study examines the independent and joint associations of physical activity (PA) and sleep with ADHD prevalence and severity.
Methods: Data from the 2016–2023 National Survey of Children's Health were used in this population-based, cross-sectional study. All measures were parent-reported. PA was categorized by weekly guideline-meeting days (≥60 min/day). Sleep duration, regularity, and ADHD severity were classified as short/optimal/long (age-specific), four-level, and three-level, respectively. Weighted multivariable logistic and ordinal regression models were used to examine the associations of PA and sleep with ADHD prevalence and severity.
Results: This study analyzed 98,182 adolescents aged 13–17, including 13,258 with ADHD. PA, sleep duration, and sleep regularity were independently associated with ADHD (all $P < 0.05$). Significant interactions between PA and sleep duration/regularity were observed for ADHD prevalence ($P < 0.05$), but not for ADHD severity. Compared to children with long sleep and 0 weekly PA days, those with optimal sleep and ≥ 1 weekly PA days had lower ADHD odds, with the lowest in the "optimal sleep and daily PA" group (OR = 0.32, 95% CI:0.21–0.51). Better sleep regularity plus more PA-meeting weekly days was linked to lower ADHD odds, with the lowest in the "always regular sleep and daily PA" group (OR = 0.25, 95% CI:0.18–0.36).
Conclusion: Even occasional engagement in regular PA was associated with lower odds of ADHD in children. Improving sleep regularity is as important as ensuring optimal sleep duration. Combining sufficient PA with healthy sleep habits showed the strongest association, underscoring the value of an integrated lifestyle approach to understanding childhood health risks.

1. Background

Attention-deficit/hyperactivity disorder (ADHD) represents one of the most prevalent neurodevelopmental disorders in childhood, affecting an estimated 2%–7% of children worldwide[49,52], with evidence of a rising prevalence[48]. ADHD in childhood predicts the subsequent development of psychiatric disorders as well as impairments in academic, occupational, and social functioning, and in adulthood it is further associated with reduced life expectancy, decreased work productivity, and substantial socioeconomic burden[1,20,37]. Although medication and behavioral interventions have demonstrated some evidence-based support in alleviating ADHD symptoms[41,57], their limitations remain significant, including limited efficacy, short duration of effect, and potential side effects[4,50]. These limitations further underscore the necessity of exploring interventions targeting modifiable risk factors and highlight the importance of proactively managing such factors in the prevention and treatment of ADHD.

In recent years, physical activity (PA) has gained attention as a non-pharmacological intervention to improve cognitive and mental health outcomes in children with ADHD[17,57]. PA is defined as any bodily movement that increases energy expenditure above resting levels[9]. The World Health Organization (2020) recommends that children and children engage in at least 60 min of moderate-to-vigorous PA daily, primarily aerobic in nature[6]. Compared to their non-ADHD peers, children with ADHD are less likely to meet the recommended 60 min of MVPA per day, and they generally engage in lower levels of PA. A nationwide prospective study has shown that lower levels of physical health, particularly in terms of cardiovascular and muscular fitness, are independently associated with a higher risk of ADHD in children and adolescents[12]. These findings support the notion that promoting PA

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and improving fitness may be reliable ways to prevent ADHD. Furthermore, an increasing body of research suggests that modifiable factors such as PA, ST, and SLP are positively correlated with mental health and social competence indicators in children with ADHD, significantly alleviating ADHD symptoms. For example, children who regularly participate in extracurricular physical activities have significantly lower levels of ADHD symptoms, with a more pronounced effect in girls[45]. Intervention studies also show that exercise can improve executive function and attention in ADHD, alleviating core symptoms[22,27,28], particularly when cognitive engagement in the activity is higher. These findings suggest that PA may play a protective role in the onset and intervention of ADHD.

Sleep is another modifiable lifestyle factor that is crucial for ADHD management, and the relationship between the roles of sleep and ADHD is interactive. Approximately 25% to 50% of children with ADHD experience sleep disorders[61], which impair attention, behavior, and daytime functioning, thereby affecting school, social, and family life [16]. Research indicates that sleep adequacy (i.e., sleep duration with age-recommended sleep duration[23] and bedtime regularity[43] are critical factors influencing ADHD. However, prior studies have often failed to distinguish between "long sleep" and "short sleep," where sleep duration deviates from the recommended guidelines, a distinction that warrants further investigation. Research has shown that children with ADHD typically experience shorter sleep duration or insufficient sleep compared to healthy controls[2,34,53]. Furthermore, longitudinal studies indicate that early childhood sleep problems may serve as predictive factors for the future development of ADHD[29,63]. Insufficient sleep may be an underlying risk factor for ADHD symptoms[53,59]. For example, studies indicate that shorter sleep duration in early childhood correlates with symptoms of inattention at age 5[29]. Further research demonstrates that inadequate sleep duration is closely associated with the severity of ADHD symptoms. Insufficient sleep exacerbates ADHD symptoms such as inattention, whereas by contrast, extending sleep duration improves attention and behavioral functioning in children with ADHD[2,51]. A recent study reported that longer sleep duration (>8.5 h) was associated with a decreased risk of ADHD symptoms in children [8]. While fewer studies have explored bedtime regularity, existing research indicates that children with ADHD exhibit greater variability in bedtime compared to controls[33]. However, given the small sample sizes (n ⟨1000), larger-scale studies are needed to explore the relationship between sleep duration, bedtime regularity, and ADHD prevalence and severity.

PA and sleep are essential components of daily life and may exert joint or combined effects on health risks. Evidence from both children [25,62] and adults[11] has demonstrated these joint effects across a variety of health outcomes and all-cause mortality[5,26]. However, most existing studies focus on the association between a single factor (e.g., whether recommended levels of PA are met) and the prevalence of ADHD, while research on the relationship between this factor and ADHD symptom severity remains scarce[24,46]. More importantly, there is still a lack of in-depth analysis of the association between the interaction of multiple factors (e.g., PA and sleep duration/regularity) and ADHD prevalence and severity.

Therefore, this study aims to use large-scale cross-sectional data to examine the associations between PA, sleep, and the prevalence and severity of ADHD, and further explore their combined effects, so as to provide new research perspectives and evidence support for the preventive intervention and clinical treatment of ADHD.

2. Methods

2.1. Data source

This study utilized the combined database of the National Survey of Children's Health (NSCH) from 2016 to 2023, comprising 334,708 cases. The NSCH is a nationwide survey conducted by the U.S. Census Bureau designed to assess American children's physical and mental health status, healthcare access, and social determinants of health. The survey employed a multi-stage data collection approach: first sending screening questionnaires to households to identify whether children aged 0–17 years were present, then randomly selecting one child from each household with children, followed by completion of an age-specific topical questionnaire by a caregiver (typically a parent) familiar with the child's health status. The sampling strategy covered all 50 U.S. states and the District of Columbia, using stratified sampling to ensure representation across states, particularly in high-poverty communities. Participants could complete the survey via web, paper, telephone, or email. Informed consent was obtained from all participating children's parents or guardians. The survey design and procedures are publicly available and described in more detail in documents provided at the following website: https://www.census.gov/programs-surveys/nsch/data/datasets.html. Ethical approval was obtained for the original data collection [64]. This study is based on existing, publicly available, de-identified data, and therefore does not involve human subjects research requirements. In conducting and writing this study, we strictly adhered to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines[60] (Supplementary Table 1).

2.2. Measures

2.2.1. ADHD and ADHD severity

Parents reported on their child's lifetime and current attention deficit disorder or ADHD, "Has a doctor or other health care provider ever told you that this child has [condition]? If yes, does this child currently have the condition?" Only current psychiatric conditions were included as indicators for this study. Parents endorsing that their child had current ADHD rated the severity as 'mild' (1), 'moderate' (2), or 'severe' (3).

2.2.2. PA

PA was assessed by the question: "During the past week, how many days did this child exercise, play a sport, or participate in PA for at least 60 min?" Response options included: "0 days", "1–3 days", "4–6 days", and "every day".

2.2.3. Sleep duration

Parents reported child's weeknight sleep duration via the question: "During the past week, how many hours of sleep did this child get on most weeknights (6–17 years)?" Response options ranged from "less than 6 hours" to "11 or more hours." While the American Academy of Sleep Medicine recommends 9–12 h for ages 6–12 and 8–10 h for ages 13–18, the NSCH questionnaire design prevented accurate identification of prolonged sleep (>12 h) in younger children[47]. Therefore, only children aged 13–18 were included, with sleep categorized as optimal (8–10 h), short (<8 h), or long (>10 h).

2.2.4. Bedtime irregularity

Parents reported on their child's weeknight bedtime irregularity with, "How often does this child go to bed at about the same time on weeknights?" Parents rated this item using the following response options: 'always' (1), 'usually' (2), 'sometimes' (3), 'rarely' or 'never' (4).

2.2.5. Measurements of other variables

Considering the effect of other factors on result, we collected data of potential confounding factors, which were shown to be associated with the dependent variable[19,35]. Participants reported on their child's age, sex, race, annual household income, highest level of education among reported adults, insurance status, premature birth, number of kids, BMI, ADHD medication currently and ADHD behavioral treatment. Household poverty level was categorized into four groups based on total family income relative to the U.S. Census Bureau's poverty threshold. Highest level of adult education was determined by the question "What is the highest grade or year of school you have completed?" and

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classified into four categories: "less than high school," "high school graduate," "some college/technical school," or "college degree and above." Insurance status was assessed through the question: "Is this child currently covered by any kind of health insurance or health coverage plan?" Premature birth was categorized as whether the child was born prematurely, and number of kids referred to the total number of children in the household. BMI is categorized into three classes based on percentile rankings: underweight (<5th percentile), healthy weight (5th–84th percentile), and overweight/obese (≥85th percentile). ADHD medication use was assessed by the question: "Is this child currently taking medication for ADD or ADHD?" Behavioral treatment was measured through the item: "At any time during the past 12 months, did this child receive behavioral treatment for ADD or ADHD, such as training or an intervention that you or this child received to help with his or her behavior, age 3-17 years?"

2.3. Statistical analysis

Following NSCH analysis guidelines, all statistical analyses in this study employed complex sampling methods with sample weights to generate accurate national estimates for the United States. Continuous variables were summarized as mean ± SD, while categorical variables were presented as frequencies with weighted percentages (%). Between-group differences were assessed using *t*-tests and χ^2 tests. Weighted binary logistic regression examined associations between PA, sleep duration, sleep regularity, and ADHD. Weighted ordinal logistic regression analyzed relationships with ADHD severity.

For independent association analyses, 0 days/week of ≥60-min PA, long sleep duration, and highly irregular sleep served as reference groups. We constructed two sequential adjustment models for our analyses. Model 1 adjusted for demographic characteristics (age, sex, race), socioeconomic factors (household income, caregiver education, insurance status), and health indicators (premature birth, number of children, BMI). Model 2 extended Model 1 by additionally adjusting for sleep duration or PA to assess their independent effects. When analyzing ADHD severity, we further adjusted for ADHD medication and behavioral treatment in addition to the covariates in Model 1.

To evaluate joint effects, interaction terms were introduced and considered significant at $p < 0.05$ (Wald test). We examined combinations of: (1) sleep duration (three categories) with PA levels (four categories); and (2) sleep regularity (four categories) with PA levels. Groups with bad sleep and 0 days/week of ≥60-min activity served as references. The joint effect analyses were based on Model 1. The same approach was applied to ADHD severity analyses within the ADHD subpopulation.

We also performed sensitivity analyses to evaluate the robustness of our findings. First, sleep duration was reclassified into two categories (suboptimal and optimal sleep) by merging participants with long sleep duration into the short sleep group due to limited statistical power in the long sleep group. Second, to address potential selection bias due to participant exclusion, we applied propensity score adjustment by including the estimated propensity score for inclusion as a covariate in the fully adjusted models. Third, inverse probability weighting analyses were conducted to account for participant exclusion, using inclusion weights combined with the study design survey sampling weights. Fourth, multiple imputation analyses were performed to address missing covariate data.

All analyses were performed using R 4.2.1, with two-sided $p < 0.05$ considered statistically significant.

3. Results

3.1. Participant characteristics

From 334,708 NSCH participants (aged 0–17 years, 2016–2023), we applied a series of exclusion criteria to refine our sample. The participant selection process is shown in Supplementary Fig. 1. The basic characteristics of included and excluded participants are shown in Supplementary Table 2. After these exclusions, the final analytic sample comprised 98,182 participants. According to ADHD status, participants were classified into ADHD (n = 13,258) and non-ADHD (n = 84,924) groups, whose baseline characteristics are summarized in Table 1. Significant differences were observed between groups in sex, race, caregiver highest education level, insurance status, premature birth, number of children in the household, and BMI (all $P < 0.05$). However, age (P = 0.40) and household poverty level (P = 0.80) did not differ significantly between the two groups.

Children with ADHD exhibited higher rates of physical inactivity (20.0% vs. 12.1%, $P < 0.001$) and lower rates of daily PA (12.8% vs. 15.1%). Additionally, ADHD participants demonstrated poorer sleep behaviors, with a greater proportion reporting irregular bedtimes (11% vs. 6.1%, $P < 0.001$) and non-optimal sleep duration (35% vs. 30%, $P < 0.001$) compared with their non-ADHD counterparts.

3.2. Association between PA, sleep and ADHD prevalence and severity

3.2.1. Association between PA and ADHD prevalence and severity

Higher levels of PA were associated with lower ADHD odds (See Fig. 1). Compared with children reporting 0 days per week of ≥60-min PA, those engaging in PA 7 days per week had 54% lower odds of ADHD (Model 1: OR = 0.45, 95% CI: 0.34–0.52; Model 2: OR = 0.46, 95% CI: 0.40–0.53). Intermediate PA levels also showed inverse associations with ADHD, with 4–6 days per week (Model 2: OR = 0.54, 95% CI: 0.47–0.62) and 1–3 days per week (Model 2: OR = 0.66, 95% CI: 0.59–0.73) both associated with lower odds of ADHD compared with inactivity. As shown in Supplementary Fig. 2, 1–3 days/week of PA (Model 2: OR = 0.73, 95% CI: 0.60–0.88) and 4–6 days/week of PA (Model 2: OR = 0.73, 95% CI: 0.56–0.96) were associated with lower odds of greater ADHD severity, whereas daily PA (7 days/week) was not significantly associated (Model 2: OR = 0.92, 95% CI: 0.71–1.18).

3.2.2. Association between sleep duration and ADHD prevalence and severity

As shown in Table 1, the independent associations between sleep duration and ADHD prevalence and severity are presented. Compared with long sleep, optimal sleep was significantly associated with lower odds of ADHD in both models (Model 1: OR = 0.54, 95% CI: 0.41–0.70; Model 2: OR = 0.57, 95% CI: 0.43–0.75). Similarly, short sleep duration showed inverse associations with ADHD (Model 1: OR = 0.69, 95% CI: 0.53–0.91; Model 2: OR = 0.72, 95% CI: 0.54–0.95). However, when examining ADHD severity among affected individuals, neither optimal sleep (Model 2: OR = 0.81, 95% CI: 0.48–1.36) nor short sleep (Model 2: OR = 0.83, 95% CI: 0.49–1.40) showed significant associations with greater ADHD severity compared with long sleep duration.

3.2.3. Association between sleep regularity and ADHD prevalence and severity

For sleep regularity, more consistent bedtime routines were associated with lower odds of ADHD. Compared with children who rarely or never had regular bedtimes, those who always maintained regular bedtimes had significantly lower odds of ADHD (Model 1: OR = 0.47, 95% CI: 0.39–0.57; Model 2: OR = 0.53, 95% CI: 0.44–0.64). Children who usually maintained regular bedtimes showed a comparable inverse association (Model 1: OR = 0.46, 95% CI: 0.39–0.53; Model 2: OR = 0.51, 95% CI: 0.44–0.59), while those who sometimes maintained regular bedtimes also had moderately lower odds of ADHD (Model 2: OR = 0.70, 95% CI: 0.59–0.83). Among children with ADHD, sleep regularity was also inversely associated with ADHD severity (Supplementary Fig. 2.). Compared with those who rarely or never had regular bedtimes, children who always maintained regular bedtimes had lower odds of greater ADHD severity (Model 2: OR = 0.59, 95% CI: 0.41–0.85), and those who usually maintained regular bedtimes demonstrated a

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Table 1
Characteristics of the study subjects ($N = 98,182$).

Variables	Overall (N = 98,182)	ADHD (N = 13,258)	No-ADHD (N = 84,924)	P Value
Age (years)	14.98 (1.41)	14.95 (1.40)	14.99 (1.41)	0.119
Sex (%)				<0.001
Boy	50,556 (50.8%)	8593 (66.1%)	41,963 (48.8%)	
Girls	47,626 (49.2%)	4665 (33.9%)	42,961 (51.2%)	
Race (%)				<0.001
Hispanic	12,499 (26.2%)	1367 (18.1%)	11,132 (27.3%)	
White, non-Hispanic	67,480 (50.3%)	9923 (60.8%)	57,557 (48.9%)	
Black, non-Hispanic	6254 (13.4%)	773 (13.6%)	5481 (13.4%)	
Other/Multi-racial, non-Hispanic	11,949 (10.0%)	1195 (7.5%)	10,754 (10.4%)	
Highest Education(%)				<0.001
Less than high school	2743 (10.9%)	269 (6.5%)	2474 (11.5%)	
High school	13,380 (19.4%)	1878 (19.7%)	11,502 (19.3%)	
More than high school	22,776 (21.9%)	3411 (24.6%)	19,365 (21.5%)	
College degree or higher	59,283 (47.9%)	7700 (49.1%)	51,583 (47.7%)	
Household poverty level (%)				0.794
0–99% FPL	10,612 (17.4%)	1642 (17.9%)	8970 (17.3%)	
100%–199% FPL	15,053 (20.8%)	2167 (20.8%)	12,886 (20.8%)	
200%–399% FPL	29,432 (29.0%)	3929 (28.3%)	25,503 (29.1%)	
≥400% FPL	43,085 (32.8%)	5520 (33.1%)	37,565 (32.7%)	
Insurance Status[a] (%)				<0.001
Insured	93,582 (92.7%)	12,797 (95.5%)	80,785 (92.3%)	
Uninsured	4600 (7.3%)	461 (4.5%)	4139 (7.7%)	
Premature birth (%)				<0.001
Yes	10,403 (10.7%)	1941 (15.3%)	8462 (10.1%)	
No	87,779 (89.3%)	11,317 (84.7%)	76,462 (89.9%)	
Number of kids[b] (%)				0.003
1	51,855 (34.3%)	6872 (34.6%)	44,983 (34.3%)	
2	32,200 (37.6%)	4377 (35.0%)	27,823 (37.9%)	
3	9986 (18.7%)	1495 (21.4%)	8491 (18.3%)	
≥ 4	4141 (9.4%)	514 (9.0%)	3627 (9.5%)	
BMI (%)				<0.001
Underweight	5300 (5.5%)	901 (6.9%)	4399 (5.3%)	
Healthy weight	67,069 (65.1%)	8238 (60.3%)	58,831 (65.8%)	
Overweight/obese	25,813 (29.4%)	4119 (32.8%)	21,694 (28.9%)	
PA (%)				<0.001
0 days	13,059 (14.6%)	2603 (19.8%)	10,456 (13.9%)	
1–3 days	40,369 (42.0%)	5710 (41.9%)	34,659 (42.0%)	
4–6 days	30,067 (27.8%)	3279 (25.4%)	26,788 (28.1%)	
7 days	14,687 (15.6%)	1666 (12.9%)	13,021 (15.9%)	
BedTime (%)				<0.001
Always	18,701 (21.1%)	2172 (19.8%)	16,529 (21.3%)	
Usually	61,848 (58.7%)	7723 (53.6%)	54,125 (59.3%)	
Sometimes	12,195 (13.6%)	2056 (15.8%)	10,139 (13.3%)	
Rarely or never	5438 (6.6%)	1307 (10.8%)	4131 (6.1%)	
Sleep duration (%)				<0.001
Optimal	69,105 (69.1%)	8721 (64.8%)	60,384 (69.6%)	
Short	27,741 (29.4%)	4265 (32.9%)	23,476 (28.9%)	
Long	1336 (1.6%)	272 (2.3%)	1064 (1.5%)	

Notes: Continuous variables are presented as mean ± SD, and categorical variables as number (weighted %). Percentages may not total 100% due to rounding. Abbreviations: ADHD = attention-deficit/hyperactivity disorder; PA, days per week with ≥60 min of physical activity; FPL = federal poverty level; BMI = Body Mass Index.

[a] Insured refers to endorsement of public (government assistance) and/or private (privately purchased, including through ACA marketplace, through employer, or TRICARE) insurance.

[b] Values represent responses from four categorical options (1 = 1 child; 2 = 2 children; 3 = 3 children; 4 = 4with children).

comparable inverse association (Model 2: OR = 0.53, 95% CI: 0.40–0.71).

3.3. Association of PA and sleep joint with and ADHD prevalence

After incorporating interaction terms into Model 1, significant joint associations of sleep duration with PA and sleep regularity with PA were observed for ADHD prevalence ($P < 0.001$). However, none of these interactions reached statistical significance in the analysis of ADHD severity. Therefore, joint association analyses were conducted only for ADHD prevalence.

The sample-weighted ADHD prevalence across combinations of PA and sleep behaviors is presented in Fig. 2. In Panel A, long sleep combined with 0 days per week of PA showed the highest prevalence (18.0%), whereas optimal sleep with 7 days per week of PA showed the lowest (9.0%). In Panel B, rarely regular sleep combined with 0 days per week of PA had the highest prevalence (21.2%), while always regular sleep with 7 days per week of PA had the lowest (8.3%). Across both panels, ADHD prevalence consistently decreased with increasing PA frequency and better sleep patterns (optimal duration or greater regularity), indicating inverse associations between these lifestyle behaviors and ADHD prevalence.

3.3.1. Association of sleep duration and PA joint with ADHD prevalence

As shown in Fig. 3., the joint associations of sleep duration and PA with ADHD prevalence are presented. Compared with children reporting long sleep and 0 days per week of PA (reference group), those with optimal sleep and 7 days per week of PA showed the lowest odds of ADHD (OR = 0.32, 95% CI: 0.21–0.51, $P < 0.001$), followed by those with optimal sleep and 4–6 days per week of PA (OR = 0.40, 95% CI: 0.25–0.61, $P < 0.001$).

Within both the short-sleep and optimal-sleep groups, engaging in as little as 1–3 days per week of PA was already associated with significantly lower odds of ADHD, with the lowest odds observed in the combination of optimal sleep and 7 days/week PA (OR = 0.32, 95% CI: 0.21–0.51), followed by optimal sleep with 4–6 days/week PA (OR = 0.40, 95% CI: 0.25–0.61). Across all sleep duration categories, a consistent trend was observed whereby higher PA frequency was

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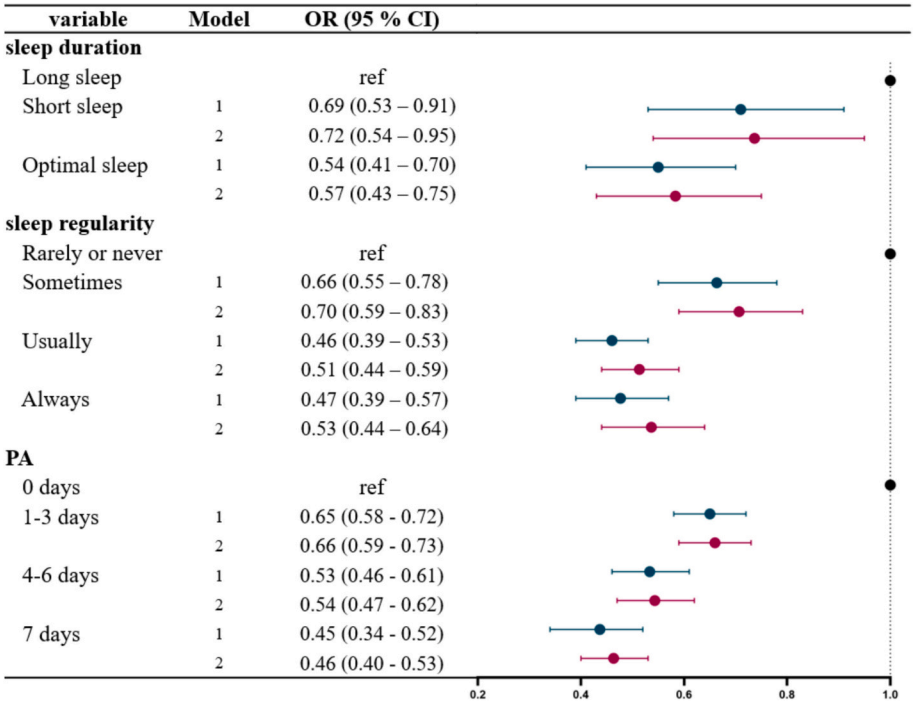


Fig. 1. Association Between Sleep, PA, and ADHD prevalence.
Note: Model 1 adjusted for age, sex, race, household income, caregiver education, insurance status, premature birth, number of children, and BMI. Model 2 additionally adjusted for Model 1 variables plus sleep duration or PA.
Abbreviations: OR = odds ratio; CI = confidence interval; PA = days per week with ≥60 min of physical activity.

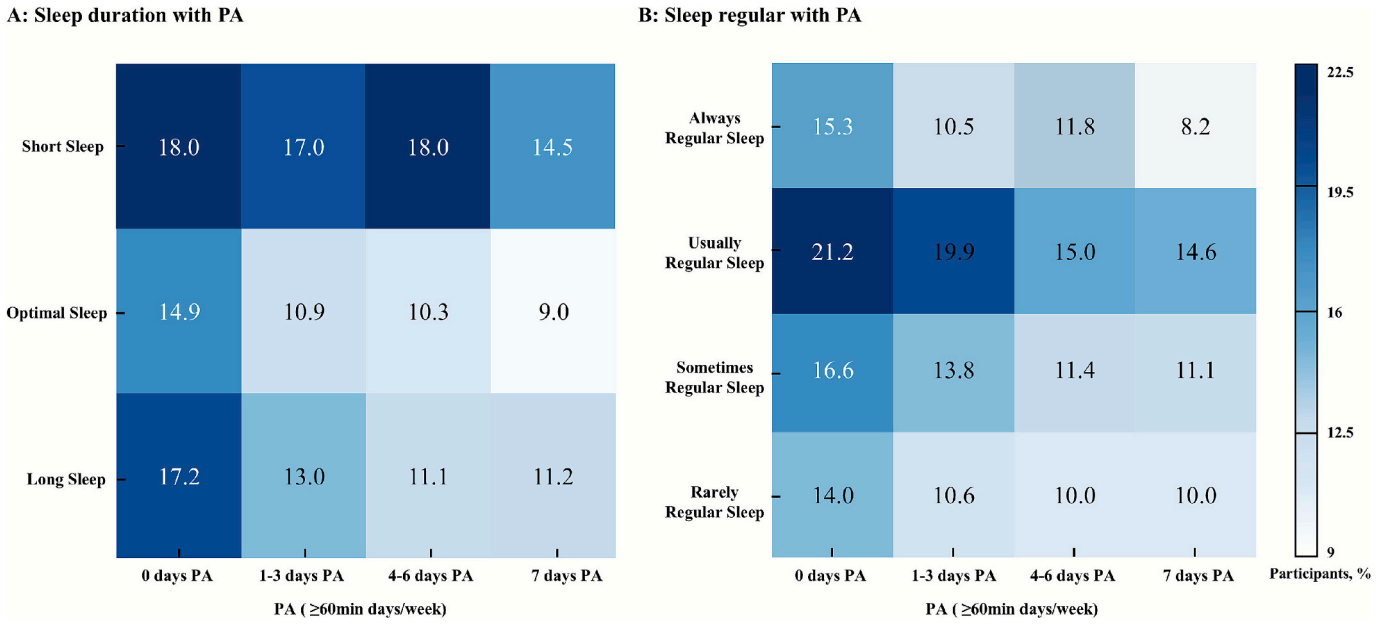


Fig. 2. Sample weighted prevalence of ADHD based on sleep-PA combinations.
Note. A: Prevalence by sleep duration and weekly days of ≥60-min PA. B: Prevalence by sleep regularity and weekly days of ≥60-min PA. Color gradient indicates prevalence percentage, with darker blue representing higher prevalence.

associated with lower ADHD odds. This pattern was most pronounced among children with optimal sleep.

3.3.2. Association of sleep regularity and PA joint with ADHD prevalence

The joint associations of sleep regularity and PA with ADHD prevalence are presented in Fig. 4. Compared with children reporting rarely regular sleep and 0 days per week of PA (reference group), those with always regular sleep and 7 days per week of PA had the lowest odds of ADHD (OR = 0.25, 95% CI: 0.18–0.36, $P < 0.001$), followed by those with usually regular sleep and 7 days per week of PA (OR = 0.28, 95% CI: 0.21–0.37, $P = 0.054$) and those with usually regular sleep and 4–6 days per week of PA (OR = 0.31, 95% CI: 0.24–0.40, $P < 0.001$). A consistent pattern was observed across all sleep regularity categories, with higher PA frequency associated with progressively lower odds of

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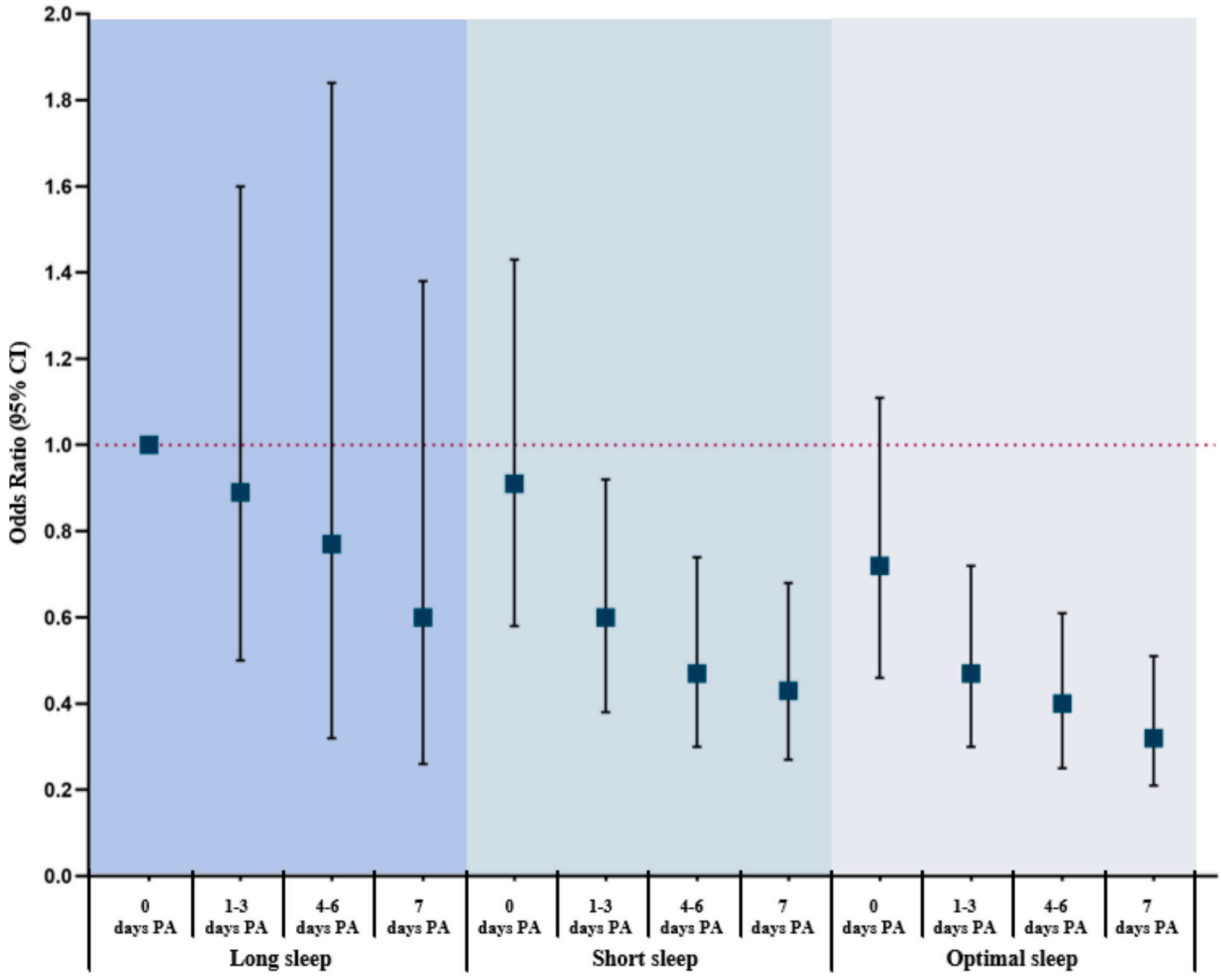


Fig. 3. Association of sleep duration and PA joint with and ADHD prevalence.
Note: Optimal sleep was defined as 8–10 h per night, short sleep as <8 h, and long sleep as >10 h.
Abbreviations: OR = odds ratio; CI = confidence interval; PA = days per week with ≥60 min of physical activity.

ADHD, indicating an inverse association even among children with less consistent sleep regularity.

3.4. Sensitivity analyses

Sensitivity analyses demonstrated that the primary associations were robust across analytical approaches. When sleep duration was reclassified into two categories—suboptimal and optimal sleep by merging participants with long sleep duration into the short sleep group—results were consistent with the main analyses (Supplementary Table 3 and 4). Similar results were obtained using the propensity score adjustment method, inverse probability weighting accounting for participant exclusion, and multiple imputation analyses addressing missing covariate data (Supplementary Table 5–8).

4. Discussion

This study revealed independent associations between higher weekly PA frequency, optimal sleep duration, and greater sleep regularity with lower ADHD prevalence. In addition, both PA frequency and sleep regularity were inversely associated with ADHD severity. Further analyses identified significant joint associations of PA with both sleep duration and sleep regularity for ADHD prevalence, although these interactions were not significant for ADHD severity. Specifically, within the PA–sleep combinations, adequate levels of either behavior appeared to attenuate the higher odds of ADHD associated with insufficient levels of the other. These findings suggest that PA and sleep behaviors are jointly associated with ADHD prevalence, highlighting the value of considering their combined influence in future research and in the broader context of child health.

Our analysis demonstrated a significant inverse association between PA and ADHD prevalence in children. Compared with children who did not meet the PA guideline on any day, those who met the recommendation of at least 60 min of PA on one or more days per week had lower odds of ADHD, with the inverse trend strengthening as PA frequency increased, and even 1–3 days per week showed potential benefits. These findings extend previous evidence. For instance, Ning Pan et al.[46] reported that children who did not meet the recommended PA level on any day had a 12% higher risk of ADHD. Regarding ADHD severity, meeting the PA guidelines on 1–3 days or 4–6 days per week was associated with approximately 27% lower odds of being in a higher ADHD severity category. This aligns with prior studies showing that PA is inversely related to ADHD symptom severity and can improve executive function[17,54,58]. The results raise the possibility of a non-linear association between PA frequency and ADHD severity in children; however, this inference is speculative and drawn by analogy to U-shaped patterns reported in adults[10]. Future studies using more detailed PA measurements (e.g., duration and intensity) and multidimensional ADHD assessments are needed to clarify whether such a pattern exists in children.

This study found that optimal sleep duration was associated with the lowest odds of ADHD, consistent with previous evidence across different

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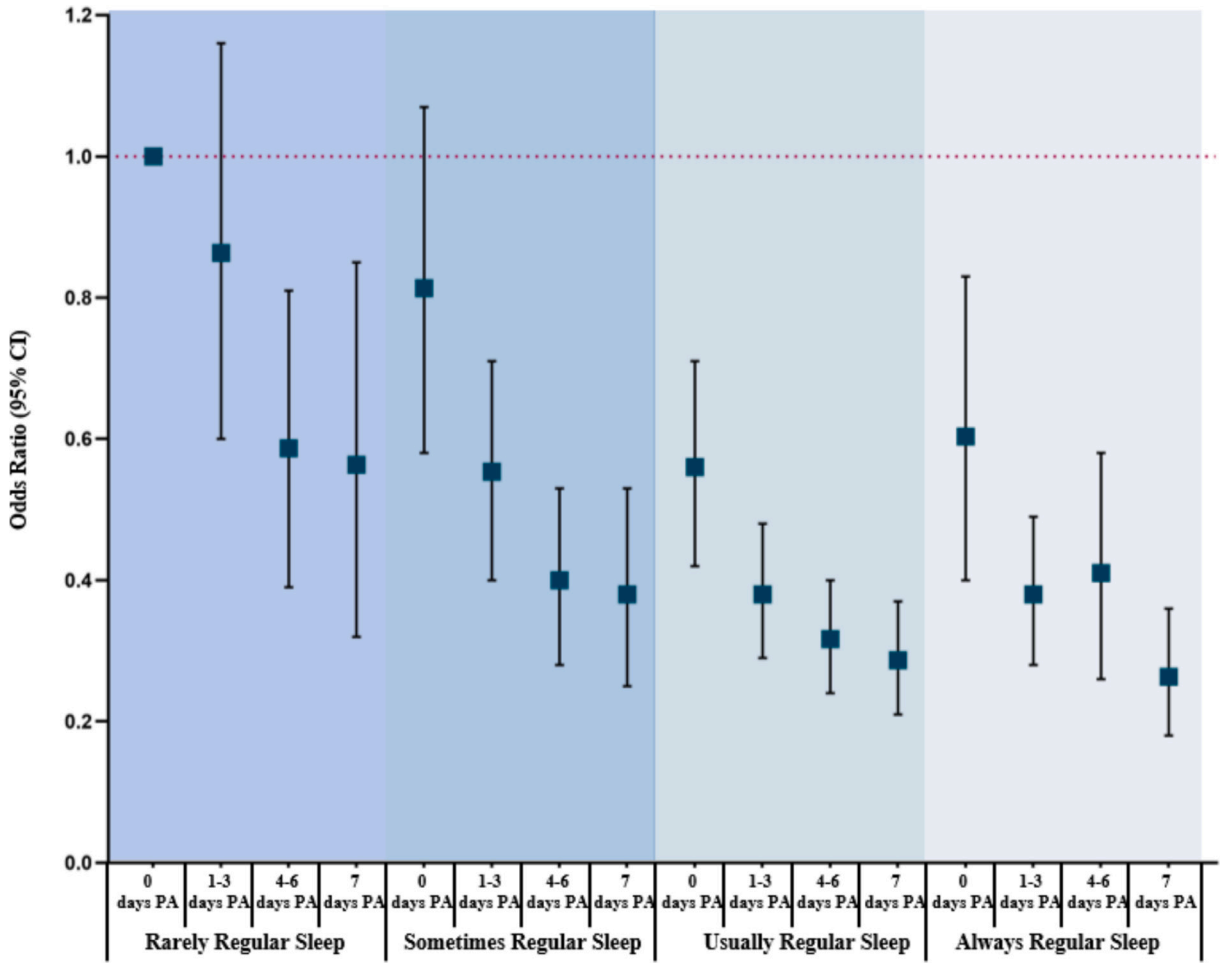


Fig. 4. Association of sleep regular and PA joint with and ADHD prevalence.

populations and health outcomes[14,21,55]. Compared with long sleep duration, short sleep was associated with lower odds of ADHD. However, This does not imply that shorter sleep may be better. This finding likely reflects characteristics captured by the long-sleep category, as previous studies have shown that children with ADHD tend to report longer sleep durations[3,7,15,30,39]. In addition, children with long sleep duration may experience prolonged sleep latency, which has been associated with greater odds of ADHD[44,56]. Given the cross-sectional design, few long sleepers, and limited sleep parameters, associations with short sleep should be interpreted cautiously. No significant association was observed between sleep duration and ADHD severity. Notably, Sleep duration in our study reflected weekday sleep only, potentially underestimating weekend compensatory sleep, consistent with a meta-analysis showing shorter sleep in ADHD primarily on weekends[40]. Additionally, the impact of sleep duration may vary across ADHD subtypes[13], underscoring the need for future research employing objective and multidimensional sleep assessments, as well as refined evaluations of ADHD symptom domains.

With respect to bedtime, previous studies have primarily focused on sleep timing rather than regularity[33]. Our findings suggest that bedtime regularity may be more strongly associated with both ADHD prevalence and severity than sleep duration. Maintaining regular bedtimes was associated with lower odds of ADHD, and the inverse association strengthened with increasing regularity. However, the estimates for the "always regular" and "usually regular" groups were nearly identical, suggesting a potential ceiling effect in the association between bedtime regularity and ADHD. Importantly, sleep regularity in this study reflects broader family and behavioral patterns rather than biological sleep mechanisms alone, making it a more modifiable factor.

Extensive evidence indicates that sleep and PA exert synergistic influences on multiple health outcomes, including mental health[25,62], cardiovascular disease[26], and all-cause mortality[36]. To our knowledge, this study is the first to systematically examine the combined associations of PA and sleep behaviors with ADHD prevalence in a pediatric population. Our results showed that the inverse association between PA and ADHD was most evident among children with optimal or short sleep duration, whereas no significant association was observed in the long-sleep group. Moreover, the elevated odds of ADHD associated with PA (0 days per week) persisted even among those with optimal sleep duration. These findings are consistent with previous experimental studies and time-substitution models, which indicate that PA may partially compensate for the adverse health correlates of inadequate sleep[36,62]. Given that modifying sleep duration can be challenging, improving sleep regularity may represent a more feasible behavioral target. In our study, children with more regular bedtimes exhibited substantially lower odds of ADHD compared with those with irregular bedtimes. Furthermore, higher PA frequency was associated with lower odds of ADHD even among children with irregular sleep patterns. Another notable finding was that within any level of sleep regularity, engaging in PA 4–6 days per week was associated with a similar reduction in ADHD odds as daily activity (7 days per week), suggesting a potential threshold effect. These results indicate that maintaining moderate-to-high PA frequency across the week, rather than daily exercise, was associated with similarly lower odds of ADHD. This pattern may have implications for promoting achievable, consistent PA routines in efforts to support behavioral and mental health in children.

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This study did not detect a significant joint effect of PA and sleep on ADHD severity. A substantial body of evidence indicates that ADHD is clinically heterogeneous, with systematic differences across subtypes in symptom presentation, comorbidity patterns, cognitive function, and neuropsychological underpinnings[38,65]. For instance, the combined subtype is often associated with more severe cognitive impairment and substance misuse[18], whereas the predominantly inattentive subtype is closely linked to social anxiety[32]. Given this heterogeneity, relying on a single global severity score may not adequately capture the potential relationships between PA, sleep, and specific ADHD symptom dimensions. Additionally, parents often lack a frame of reference for their own child's behavior, parent ratings may be influenced by treatment, expectations, or caregiver burden[31].

Strengths of this study include the integration of data from multiple survey cycles, yielding a large, nationally representative sample of U.S. children and adolescents. Sample-weighted and multivariable-adjusted analyses were applied across all regression models to enhance the reliability and robustness of the findings. Furthermore, This study examined sleep duration and regularity together to better assess their associations with PA and ADHD. However, this study also had several limitations. First, cross-sectional analyses are limited to describing associations, not proving causation. It is also possible that ADHD symptoms influence both sleep and PA. Second, reliance on parent-reported measures may introduce measurement error[42]. Third, PA and sleep variables were categorical, limiting the ability to examine dose-response relationships or identify precise behavioral thresholds. Future research should integrate multi-informant and multidimensional ADHD assessments (e.g., teacher reports, executive function tests), employ objective sleep monitoring (e.g., actigraphy), and adopt longitudinal designs to clarify temporal relationships and better understand these interactions.

5. Conclusion

This cross-sectional study provides valuable insights into the associations of PA and sleep habits with ADHD. Meeting PA recommendations and maintaining healthy sleep (including adequate duration and regular bedtime) were independently associated with lower ADHD prevalence and milder symptoms, with partially overlapping effects. From a public health perspective, supporting children in improving multiple behaviors concurrently while recognizing the value of partial changes may hold practical relevance. These lifestyle factors serve as supportive adjuncts rather than substitutes in ADHD management, a role that requires further validation through longitudinal or mechanistic research.

CRediT authorship contribution statement

Beibei Shi: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Hong Mou:** Writing – review & editing, Conceptualization. **Zhiduo Chen:** Writing – review & editing, Methodology. **Changshuang He:** Writing – review & editing, Conceptualization. **Meng Zhang:** Writing – review & editing, Conceptualization. **Zimeng Zhang:** Writing – review & editing. **Huakun Zheng:** Writing – review & editing. **Jing Wang:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Minghui Quan:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors confirm no potential conflicts of interest, either financial or personal, that might have biased the outcomes presented in this study.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.genhosppsych.2026.01.016.

Data availability

The datasets analyzed during the current study are available in the NSCH dataset repository, https://www.census.gov/programs-surveys/nsch/data/datasets.html.

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